Admin Building Weekend Dip

Overview
Admin Building Weekend Dip is an energy analytics project that identifies reduced electricity consumption patterns during weekends in administrative buildings. The system applies k-means clustering on daily usage profiles to automatically separate weekday and weekend behavior. Regression models are then used within each cluster to forecast future consumption, and savings potential is visualized through pie charts on a Streamlit dashboard to support energy optimization decisions.

Problem Statement

Administrative buildings operate at high capacity during weekdays but often continue consuming unnecessary electricity during weekends due to partially active HVAC, lighting, and equipment. The objective is to detect weekend consumption dips, forecast future usage patterns, and estimate potential energy savings to enable efficient operational planning.

Dataset
The project uses the Electricity Load Diagrams 2011–2014 dataset containing 15-minute smart meter readings for multiple clients. One meter is treated as an administrative building. The readings are converted from kW to kWh and aggregated into daily usage profiles to analyze weekday and weekend consumption differences.

Workflow
Load the dataset and convert the timestamp column to datetime. Convert 15-minute readings into kWh and aggregate electricity consumption at daily level. Create additional features such as day of week and weekend indicator to distinguish operational patterns. Apply k-means clustering on daily electricity usage to group similar behavior patterns such as weekday high usage and weekend low usage. Train regression models within each cluster to forecast future electricity demand. Compare weekday and weekend averages to compute savings potential. Visualize clusters and savings through scatter plots and pie charts on the dashboard.

Technologies Used

Python, Pandas, Scikit-learn (K-Means and Linear Regression), Plotly, Streamlit.

Installation
Install dependencies using: pip install pandas scikit-learn plotly streamlit

How to Run

Place the Electricity Load dataset file in the project directory and run: streamlit run app.py

Key Features

Discovers usage patterns automatically using clustering, performs cluster-based forecasting, estimates weekend energy savings, and provides interactive visualization of consumption behavior.

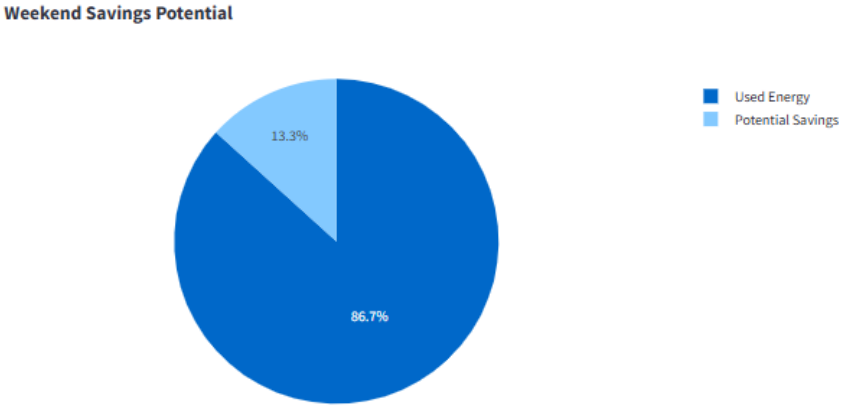
Applications
Administrative building energy optimization, smart campus management, weekend shutdown planning, sustainability reporting, and cost reduction strategies.

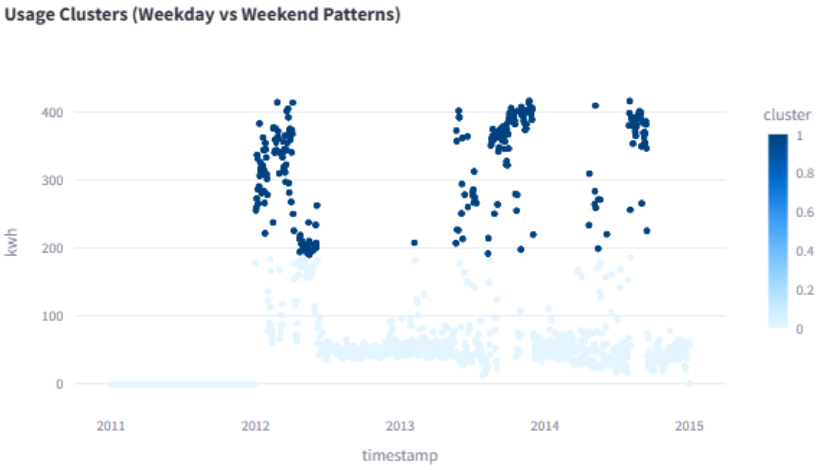
Limitations
Only one smart meter is used as a proxy for an administrative building, clustering is based mainly on

usage magnitude, and external factors such as occupancy schedules and weather conditions are not included.

Output

The dashboard displays clustered usage patterns highlighting weekday and weekend behavior along with a pie chart showing the share of weekend consumption versus potential savings, enabling clear identification of optimization opportunities.





Future Enhancements

Integrate occupancy and calendar data, include weather variables, apply advanced clustering methods, implement automated energy-saving recommendations, and enable real-time monitoring.

Conclusion

The project demonstrates that k-means clustering combined with cluster-specific regression can effectively identify weekend electricity dips in administrative buildings and quantify potential savings, enabling informed decisions for energy-efficient building operations.